

COLLEGE ALUMNI CONNECT & MANAGEMENT SYSTEM

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Abstract— Alumni relationship management represents a critical challenge for higher educational institutions in sustaining long-term engagement, career support, and institutional development. Weak communication between alumni, students, faculty, and administrators often results in underutilization of alumni expertise, reduced mentorship opportunities, and fragmented data management. Traditional alumni management practices are largely manual, subjective, and inefficient, making them unsuitable for modern, large-scale academic environments. Moreover, existing digital alumni platforms primarily focus on data storage and event notifications without intelligent matchmaking or personalized engagement mechanisms, thereby limiting their effectiveness.

To address these challenges, this project proposes a comprehensive College Alumni Connect & Management System that integrates role-based portals with an AI-based Skill Matching Module to enhance alumni–student interaction and mentorship accuracy. The proposed system preprocesses and analyzes student skills, interests, and career goals, and intelligently matches them with suitable alumni mentors using machine-learning-driven similarity analysis. Dedicated portals for students, alumni, faculty, and administrators ensure secure communication, efficient data handling, and systematic monitoring of engagement activities.

The effectiveness of the system is validated through functional evaluation of mentorship accuracy, communication efficiency, and system usability. Experimental observations indicate that the AI-based skill matching significantly improves mentor relevance and reduces manual intervention.

The system not only facilitates mentorship and career guidance but also supports alumni networking, job sharing, feedback management, and institutional reporting. Overall, the proposed system demonstrates strong potential as a scalable, intelligent, and practical solution for strengthening alumni engagement and student career development in higher education institutions.

Keywords— Alumni Management System, AI-based Skill Matching, Mentorship Platform, Educational Data Management, Student–Alumni Engagement.

Introduction

Alumni play a vital role in shaping the academic reputation, career outcomes, and long-term sustainability of educational institutions. Through mentorship, career guidance, industry exposure, and institutional contributions, alumni significantly influence student success and organizational growth. However, maintaining consistent and meaningful engagement with alumni remains a persistent challenge for many institutions worldwide.

In many colleges and universities, alumni data is scattered across spreadsheets, legacy systems, or informal communication channels, leading to inefficient information management and minimal interaction. Manual methods of tracking alumni achievements, organizing events, and coordinating mentorship are time-consuming, subjective, and often unreliable, particularly in institutions with large alumni networks. As a result, students often miss valuable guidance opportunities, while alumni remain disconnected from their alma mater.

With advancements in web technologies and artificial intelligence, there is an increasing demand for intelligent platforms that go beyond basic data storage to provide personalized engagement and decision support. AI-driven recommendation systems have shown promising results in domains such as e-commerce and education, particularly in matching users based on preferences, skills, and historical data. However, their adoption in alumni management systems remains limited.

One of the key challenges in alumni–student mentorship is identifying the most suitable mentor for a given student. Manual mentor selection often results in mismatches due to limited information and human bias. This issue is further compounded by diverse academic backgrounds, career paths, and skill requirements. An automated and intelligent mentor-matching mechanism is therefore essential for improving mentorship outcomes.

The proposed College Alumni Connect & Management System addresses these issues by providing a centralized, role-based platform integrated with an AI-based Skill Matching Module. The system enables secure communication, structured mentorship, alumni networking, feedback management, and institutional reporting. By leveraging artificial intelligence, the system enhances mentor relevance, improves engagement efficiency, and supports data-driven decision-making for administrators.

Literature Survey :

Several studies and existing systems have explored alumni management and student–alumni interaction through web-based platforms that primarily focus on storing alumni information, sharing job opportunities, and announcing events. Traditional alumni systems are mostly static and lack real-time communication, personalized mentorship, and intelligent recommendation features. Some mentorship and career guidance platforms attempt to connect students with professionals, but they rely heavily on manual mentor selection, making the process inefficient and less effective. Recent research in recommendation systems highlights the use of

skill-based matching techniques to improve relevance and user satisfaction; however, such AI-based approaches are rarely integrated into academic alumni platforms. Additionally, existing systems often fail to provide role-based access, centralized administration, and structured feedback management. These limitations indicate a clear need for an integrated, intelligent, and scalable solution, which motivates the development of the proposed College Alumni Connect & Management System with AI-based skill matching to enhance mentorship, communication, and long-term alumni engagement.

RELATED WORK

Several alumni management platforms and educational networking systems have been developed to improve institutional engagement. Early alumni systems primarily focused on maintaining contact information and sending event notifications. While these systems improved data organization, they lacked advanced interaction and personalization features.

Recent studies highlight the importance of digital alumni portals with secure authentication and role-based access control. Some systems incorporate basic analytics and reporting; however, most fail to provide intelligent mentorship recommendations. Unlike learning management systems, alumni platforms often overlook AI-driven personalization.

Recommendation systems based on skills and interests have been successfully applied in academic advising and job portals. However, their integration into alumni management remains limited. Existing platforms often rely on manual mentor assignment, which is inefficient and prone to bias.

Furthermore, many systems emphasize system functionality but neglect usability, long-term engagement, and decision-support capabilities. Comparative analysis of AI models and their impact on mentorship accuracy is rarely addressed. The proposed system overcomes these limitations by integrating AI-based skill matching

with structured alumni management and practical engagement tools.

PROPOSED METHODOLOGY

The proposed system follows a modular architecture consisting of Student Portal, Alumni Portal, Faculty Portal, Admin Panel, and AI-based Skill Matching Module. User authentication ensures secure access based on roles.

The AI module analyzes student profiles and alumni profiles using similarity metrics derived from skills, interests, and career objectives. Matching scores are computed, and the most relevant alumni mentors are recommended to students. Automated notifications are then generated to initiate mentorship interactions.

Unlike traditional systems that rely on manual matching, the proposed methodology integrates intelligent recommendation directly into the alumni management workflow, ensuring efficiency and scalability.

Data Collection & Preprocessing

The system collects data through structured registration forms for students, alumni, faculty, and administrators. Student data includes academic background, skills, interests, and career goals, while alumni data includes professional experience, industry domain, skills, and mentorship preferences. Faculty data focuses on academic oversight and recommendations, while administrators manage system-level configurations.

All collected data is securely stored in a centralized database with role-based access control to ensure privacy and integrity. Preprocessing includes data normalization, skill categorization, and removal of incomplete or redundant entries. Historical interaction data is also utilized to improve mentor recommendation accuracy.

The preprocessing stage ensures that the AI-based Skill Matching Module receives clean, standardized, and relevant input data, thereby enhancing recommendation performance.

IMPLEMENTATION

The system is implemented using modern web technologies with a secure backend and relational database. The AI-based Skill Matching Module utilizes machine learning techniques to calculate similarity scores and refine recommendations over time.

Administrators can monitor system activity, generate reports, manage users, and configure notification rules. Faculty members can track mentorship effectiveness and recommend students for alumni guidance.

The modular implementation ensures extensibility, allowing future integration of mobile applications, advanced analytics, and additional AI models

RESULTS & DISCUSSION

The system demonstrated improved mentorship accuracy, faster communication, higher engagement levels compared to traditional alumni management approaches. AI-based recommendations reduced manual mentor selection and improved satisfaction among students and Alumni. Administrators benefited from centralized reporting and real-time monitoring, while faculty gained better visibility into student–alumni interactions. Overall, the system proved effective in strengthening alumni relationships and supporting student career development.

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CONCLUSION

The College Alumni Connect & Management System provides an intelligent, secure, and scalable solution for modern alumni engagement. By integrating AI-based skill matching with structured role-based portals, the system enhances mentorship accuracy, improves communication efficiency, and supports institutional growth. Although future enhancements such as mobile deployment and advanced analytics can further improve performance, the proposed system demonstrates strong potential for practical implementation in higher education institutions.

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